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Towards Prediction of Landslide Susceptibility using Random Forest for Kalutara District, Sri Lanka

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***Abstract*—Landslide susceptibility prediction has been a topic of interest in the world of research for many years. The prime objective of this study is to explore the integration of geospatial techniques such as Geographic Information Systems(GIS) with machine learning techniques to get refined and improved models with better landslide prediction accuracy. In this paper, a comprehensive study on landslide susceptibility in Kalutara district, Sri Lanka was carried out for a total of 84 landslide occurrences under 12 conditioning factors; slope, aspect, geology, hydrology, soil type, soil thickness, land use, landform, Sediment Transport Index(STI), Topographic Wetness Index(TWI), Stream Power Index(SPI), and rainfall. The results obtained by analyzing the 12 landslide conditioning factors using the Frequency Ratio and Information Gain Ratio are used to construct a landslide susceptibility prediction model using the Random Forest algorithm.**

***Index Terms*—Landslide Susceptibility, Machine Learning, GIS, Random Forest**

I. Introduction

Landslides, simply the mass movements of the earth surface, is a phenomenon of displacement of soil, rock or debris along the slope by mainly due to gravitational forces [1]. Of the 65,000 sq km of the land extent in Sri Lanka, an area of nearly 20,000 sq km [1] encompassing 10 districts is identified as prone to landslides. Such an area covers approximately 30% of land area in Sri Lanka. As per statistics obtained by the National Building Research Organization(NBRO), Sri Lanka, more than 400 lives have been lost due to landslides during the past three years in Sri Lanka. In 2016 and 2017, 151 and 230 [2] human lives were lost due to landslides respectively, with an estimated 10,000 families being displaced. 101 deaths among them were reported in Kalutara district. Owing to haphazard, unplanned land use, inappropriate construction methods, wanton human intervention and other geological and morphological causes, the trend of landslide occurrence will likely continue in the next eras. Therefore, understanding landslide susceptible areas and prediction of landslide susceptibility with high accuracy play an indispensable role in planning risk mitigation actions and related decision making.

Landslides can be a result of a broad variety of landslide conditioning factors [3]. However, only certain classes of conditioning factors will have a considerable impact on landslide occurrence [4] in a given study area. In this study, the relationship between landslide conditioning factors and landslide occurrences has been analyzed [5] since the selection of the best set of conditioning factors for the study area is identified as crucial [6].

Highly developed Geographic Information Systems together with mathematical and machine learning algorithms, have enabled effective landslide modeling [8]. The confusion on which techniques or models will predict the landslide susceptibility with high accuracy [9] still prevails. A suitable high performing landslide susceptibility model is expected to demonstrate a rise in the prediction accuracy of about 1 or 2% [10] when compared to other models.

This paper presents an analysis of landslide conditioning factors using Frequency Ratio(FR) and selection of conditioning factors using Information Gain Ratio towards the prediction of landslide susceptibility using Random Forest(RF).

The rest of the paper is organized as follows. Section II provides an analysis of the previous literature. Section III includes a description of the area selected for the study. Section IV discusses the suggested methodology of the study and Section V outlines the preliminary results of the study.

II. Related Work

Landslide susceptibility mapping approaches are classified into two as qualitative or quantitative approaches [8]. Qualitative methods usually depend on expert opinions [11] whereas quantitative methods depend on the relationships between landslide controlling factors and landslides [12]. Figure 1 demonstrates a classification of landslide susceptibility mapping methods.

Qualitative studies [13], [14] consider a set of conditioning factors and based on the importance, factors are weighted using methods like Analytic Hierarchy Process(AHP). Nevertheless, quantitative methods have become widely used in recent years [12]. Frequency Ratio (FR) [16],[17],[18] and Logistic Regression (LR) [15] are usually employed in landslide susceptibility mapping, even though a variety of other statistical methods exist. These two methods can perform bivariate statistical analysis(BSA) and multivariate statistical analysis(MSA) respectively. To effectively overcome the limitation of data-dependent bivariate and multivariate statistical methods, machine learning-based models can be used.

C. Zhou et al. [15] have used Logistic Regression(LR), Artificial Neural Network(ANN), Support Vector Machine(SVM), and in landslide susceptibility modeling in Three Gorges Reservoir area, China. Information Gain Ratio was used to

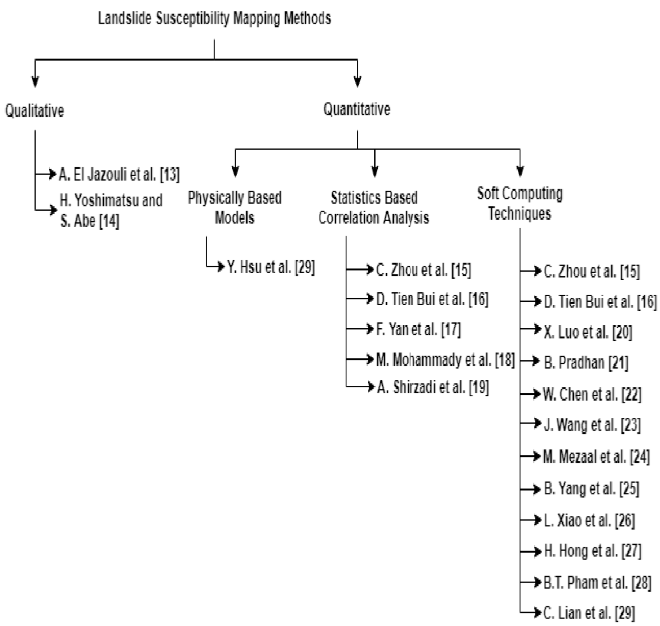

Fig. 1. Classification of Landslide Susceptibility Mapping Methods

select the best subset of landslide conditioning factors out of 12 initial factors. When the performance was evaluated using the Friedman test and ROC curve, the SVM model showed an Area Under Curve(AUC) of 0.937 and 0.912 on training and testing data respectively while the LR model showed an average performance. SVM and ANN models outperformed the LR model while SVM was found the ideal for the study area.

X. Luo et al. [19] evaluated a landslide inventory database of 493 landslides under 16 conditioning factors using ANN, SVM, and one statistical model, Information Value Model (IVM). ANN model achieved higher prediction capability than SVM and IVM. Biswajeet Pradhan [20] compared the performance of Decision Tree(DT), SVM, and Adaptive Neuro-Fuzzy Inference System (ANFIS) where ANFIS showed a high success rate(81.46) and prediction rate(82.80).

Landslide susceptibility modeling results obtained from a study conducted by Wei Chen et al. [21] for 222 landslide locations in Chongren County, China using LMT, Bayes' Net(BN), Random Forest(RF) and Radial Basis Function(RBF) Classifier showed that the Random Forest(RF) model as having the highest specificity, sensitivity, and accuracy values of 0.716, 0.787, and 0.752 respectively for the training data set.

Another study conducted by Wei Chen et al. [22] compared the landslide susceptibility predictive ability of Classification and Regression Tree (CART), Random Forest, and LMT models under 12 landslide conditioning factors which revealed positive predictive abilities when validated using Linear Support Vector Machine(LSVM) method. Each model performance when compared against values of ROC curves, Std. error, ACC, AUC, CI at 95%, and significance level P, indicated that the Random Forest model having the highest prediction rate of 0.781.

A neural architecture for automatically detecting landslides using airborne, high-resolution laser scanning data was proposed by M. Mezaal et al. [23]. A total of 21 landslides were analyzed using Recurrent neural networks (RNN) and multi-layer perceptron neural networks (MLP-NN) in landscape detection based on high-resolution LiDAR data. RNN and MLP-NN gave out 83.33% and 74.56% respectively, as test accuracy results.

The study of Beibei Yang et al. [24], presented a model to predict the displacement of landslides using the Long-Short Term Memory network (LSTM) and time series analysis. When the success or failure of the model was compared with Baishuihe and Bazimen landslide inventory, LSTM based model provided 7.11 RMSE (Root Mean Squared Error) value while SVM based model provided 21.83 RMSE value, indicating LSTM based prediction model has outperformed surpassing SVM based classifications. The prediction accuracy of SVM, LSTM, Back Propagation Neural Network(BPNN), and DT in the study conducted by Liming Xiao et al. [25] were 62.0%, 72.9%, 60.4% and, 81.2% respectively. It was found that the model based on LSTM performed with sufficient accuracy because of its capability to learn time series based patterns along with temporal dependencies.

Haoyuan Honga et al. [26] explored the use of ensemble techniques, Rotation Forest, Bagging, and AdaBoost for landslide susceptibility assessment with J48 Decision Tree (JDT) in which the model based on Rotation Forest outperformed with the highest prediction capability, (AUC = 0.855). The performances of five hybrid machine learning techniques, Rotation Forest-based Reduced Error Pruning Trees(RFREPT), Multi-Boost based Reduced Error Pruning Trees (MBREPT), Reduced Bagging based Reduced Error Pruning Trees(BREPT), Reduced Error Pruning Trees(REPT), and Random Subspace-based Reduced Error Pruning Trees(RSREPT) were evaluated and the results were compared by B. Pham et al. [27]. In this study, the BREPT model had outperformed other models.

Among the quantitative approaches, physically-based approaches [28] are expensive and will not be practical for large landslides sites. Few studies [16], [29], [30] have been carried out employing both qualitative and quantitative approaches to predict landslide susceptibility.

III. Case Study: Kalutara District

The study area, Kalutara District(figure 2) is located in the Western Province, Sri Lanka within latitude 6°34′59.99″ N and longitude 80°9′60.00″ E. It covers a total area of land of 1,598 sq.km. of which inland waters account for 8.4 sq.km. The topography of the area indicates that about 50% of the district is under 200 ft, about 99% is under 1500 ft. and only about 1% is over 1500 ft [31]. The district has three major rivers namely, Kalu, Panadura, and Bentota. The average annual rainfall in the district is around 2500mm[32]. The maximum rainfall for the district is generally received during the southwest monsoonal season. Plantations such as tea, rubber, paddy, cinnamon, pepper, coconut, etc. cover a major part of the land use in the selected study area. The prominent

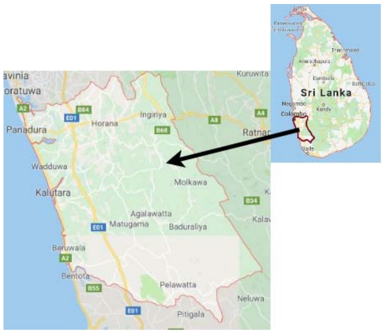

Fig. 2. Study Area:Kalutara District

lithologies of the Kalutara District include Quartzite, Granite Biotite Gneiss, Charnochite, Charnockitic Gneiss, Khondalite, Quartzo Feldspathic, and Granulatic Gneiss. The bedrock materials in the area can be covered by Alluvial, Boulders, Colluvium, Residual and Rock Exposure overburden.

No earthquake-induced landslides have been reported before the time of this study in Kalutara district. High-intensity rainfall has triggered the recorded landslides. Due to massive deforestation and inappropriate land management, the frequency of the landslides in the Kalutara district has increased during the past decade.

IV. Methodology

The study employs a hybrid approach to a quantitative research approach and a constructive research approach. The methodological flow carried out in this study consists of six main steps discussed below.

A. Landslide Inventory Mapping and Preparation of Training and Testing Data Sets

The landslide inventory map was created considering the locations where landslides occurred in the past in the study area, so as to depict the spatial distribution of the landslide events. 84 landslide occurrences were analyzed in this study. For the susceptibility analysis, 70% of the landslide locations were randomly selected for training and 30% for testing. Landslide pixels were assigned a value, 1, while non-landslide pixels were assigned a value, 0. Landslide locations denote the main scarp of the respective landslide.

B. Preparation of Landslide Conditioning Factor Maps

Based on the knowledge attained from the previous literature and expert consultation from the NBRO, 12 conditioning factors were selected for the study, including slope, aspect, hydrology, landform, land use, soil type, soil thickness, geology, Topographic Wetness Index(TWI), Stream Power Index(SPI), Sediment Transport Index(STI), and rainfall. A spatial database containing landslide related conditioning factors was constructed using data extracted from the contour map of Kalutara district, geospatial statistical data, and precipitation data acquired from the NBRO. QGIS was used to generate thematic maps for each of the conditioning factors. The conditioning factor maps were converted into a pixel format with a spatial resolution of 10×10 m.

Using contour lines, Digital Elevation Model(DEM) was created with a grid size of 10×10 m. Slope, aspect, SPI, STI, and TWI were extracted using the DEM. The slope is defined as the change in elevation with a change in the horizontal position. The slope map was reclassified into four classes. Aspect is the compass direction that a slope faces. Aspect values were divided into three classes.

SPI quantifies the erosive power of flowing water. TWI measures topographic control on hydrological processes. STI describes the process of erosion and deposition. SPI, TWI, and STI are calculated using equation 1,2 and 3 respectively.

$$SPI = Atan\left(\frac{\beta}{b}\right) \tag{1}$$

$$TWI = \log_e\left(\frac{A}{btan\beta}\right) \tag{2}$$

$$STI = \left(\frac{A_s}{22.13}\right)^{0.6}\left(\frac{sin\beta}{0.0896}\right)^{1.3} \tag{3}$$

β (radian) is the slope at a given cell, b (m) is the cell width through which water flows, A (m^2) is the flow accumulation, and A_s is the upstream area.

Hydrology indicates the distance to waterways in Kalutara district and it was divided into six classes. The natural features and shapes existent on the earth's surface are identified as the land-forms. The study area is comprised of seven types of land-forms including clay flats, clay swamps, scattered small hills, flat plains, marches, wetlands, etc. The land use varies from crops, forests to other non-agricultural uses distributed over ten classes. Alluvial, boulders, colluvium, residual, and rock exposure are the most prominent soil types in Kalutara district. The thickness of the soil in these types varies throughout the region. Geology varies withing 7 classes, namely; Quartzite, Granite Biotite Gneiss, Charnochite, Charnockitic Gneiss, Khondalite, Quartzo Feldspathic, and Granulatic Gneiss. Rainfall data during the times of landslide occurrences were gathered from the Department of Meteorology, Sri Lanka and was divided into six classes.

C. Correlation Analysis

To analyze the spatial relationship between the landslides and conditioning factors Frequency Ratio(equation 4) was used in the study. The ratio of the probabilities of presence to absence landslide occurrence [17] for a specific conditioning factor is indicated by the results obtained for frequency ratio. FR>1 suggests a higher correlation while FR<1 suggests vice versa [18]. FR=1 represents an average value.

$$FR_i = \frac{N_{pix(S_i)}/N_{pix(N_i)}}{\sum N_{pix(S_i)}/\sum N_{pix(N_i)}} \tag{4}$$

$N_{pix(S_i)}$ is the number of landslide pixels in each conditioning factor class i. $N_{pix(N_i)}$ is the total number of pixels that have class i in the study area. $\sum N_{pix(S_i)}$ is the total number of landslide pixels in the study area. $\sum N_{pix(N_i)}$ is the total number of pixels in the study area.

D. Selection of Conditioning Factors

The predictive capability of the employed model can be reduced by features with a certain noise level. Information Gain Ratio was used to quantify the predictiveness of the 12 landslide conditioning factors and eliminate the factors with zero predictiveness. This assists in achieving more accurate prediction [6].

The ratio of information gain to the intrinsic information is given by the Information Gain Ratio. Predictiveness is considered high when the information gain ratio value is high[6]. The Information Gain Ratio for landslide conditioning factor X is computed using equation 5,6,7, and 8.

$$Info(S) = -\sum_{i=1}^{2} \frac{n(L_i, S)}{|S|} log_2 \frac{n(L_i, S)}{|S|} \tag{5}$$

$$Info(S, X) = \sum_{j=1}^{m} \frac{S_j}{|S|} Info(S) \tag{6}$$

$$SplitInfo(S, X) = -\sum_{j=1}^{m} \frac{S_j}{|S|} log_2 \frac{|S_j|}{|S|} \tag{7}$$

$$Information\ Gain\ Ratio(S, X) = \frac{Info(S) - Info(S, X)}{SpiltInfo(S, X)} \tag{8}$$

The training data set is denoted by S. n(L_i,S) is the number of class L_i samples in S. Info(S) is the Information Gain. Info(S,X) is the amount of information required to split S into m subsets related to X. Information generated by splitting S into m subsets is denoted by SplitInfo.

E. Construction of Landslide Susceptibility Prediction Model

Initially, three candidate machine learning algorithms were identified through the previous literature to implement the model. They are Support Vector Machine(SVM) [15],[19],[25], Artificial Neural Networks(ANN) [15],[19], and Random Forest(RF) [21],[22],[35],[36],[37]. SVM is a supervised classifier that separates the feature space obtained by the input data set into classes, using a hyper-plane which creates the maximum margin[33]. ANNs consist of a chain of nodes called "Artificial Neurons", that are interconnected and able to identify the relationship patterns between input-output layers [34]. RF is an ensemble classifier that uses multiple decision trees to make the final prediction. Out of the identified candidate algorithms, Random Forest was selected to construct the model considering the performance [21],[22],[35],[36],[37] of the RF models in previous studies. Random Forest has also not been used in the prediction of landslides in Sri Lanka before the time of this study.

RF implements a forest of random decision trees using the training set and assigns each tree as a classification rule. RF employs a subset of predictors that are randomly selected to perform the best possible split at each node. This random feature selection reduces the correlation among two decision trees and reduces the out of bag error of any tree in the forest. RF utilizes bagging to produce ensemble predictions from a forest of trees.

F. Model Performance Evaluation

The landslide susceptibility prediction model was evaluated using Receiver Operating Characteristic(ROC) method. The area under the ROC curve(AUC) provides the overall measure of test performance. According to Tien Bui et al. [6], the correlation between AUC value and the predictive ability of a model can be decided as follows: (0.9–1) an excellent model, (0.8–0.9) very good model, (0.7–0.8) good model, (0.6–0.7) average model, and (0.5–0.6) indicates a poor model.

Furthermore, three statistical measures; accuracy, sensitivity, and specificity are used to evaluate the prediction model. Accuracy(equation 9) is the fraction of correctly classified landslide and non-landslide pixels. Sensitivity(equation 10) is a measure of true landslide pixels classified as landslide occurrences. Specificity(equation 11) is the proportion of the true non-landslide pixels classified as non-landslide occurrences.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{10}$$

$$Specificity = \frac{TN}{FP + TN} \tag{11}$$

TP(True Positive) indicate landslide instances classified as landslides. TN(True Negative) indicate non-landslide instances classified as non-landslides. FP(False Positive) and FN(False Negative) show non-landslide instances classified as landslides and landslide instances classified as non-landslides respectively.

V. RESULTS

The preliminary results acquired in the study are outlined in this section. Table I shows the values obtained from the Frequency Ratio analysis between landslides events and conditioning factors. Resulting FR values indicate that Quartzo Feldspathic geology class demonstrates a comparatively high relevance (FR = 2.29) to the landslide event than other categories of geology. Further, geology with GraniteBiotite Gneiss, Charnochite, and Khondalite areas, showed the lowest FR (FR = 0.000), which indicates a low chance of landslide occurrence. In terms of soil type, colluvium has the highest FR(2.99). Therefore, the probability of landslide occurrence with colluvium soil type is greater than that of the other soil types. The landform classes of D38 and D32 have FR values of 0.107 and 0.082 indicated a high probability of landslide occurrence compared to other landform classes. The results also showed that land use for village home-lets has the highest(FR=10.583) probability for landslide occurrence. Residual soil thickness in the range of 4-5m and colluvium soil type with 2m thickness have the highest(FR=0.597) and lowest(FR=0.024) probabilities respectively to cause a landslide event. Areas, where the distance to waterways are ranging from 300 to 600, are more susceptible to landslides with FR value 2.62. The rainfall values between 303mm-439mm showed a high probability to trigger a landslide event.

Attribute	Classes : FR			
Geology	Qtz	1.55	GtBtGn	0
	Ch	0	ChGn	1.07
	Kh	0	QtzFd	2.29
	GrGn	0.07	NM	0
Soil Type	Alluvial	2.76	Boulders	0.68
	Colluvium	2.99	Residual	1.02
	Rock Exposure	0.35		
Soil Thickness	coll t=1	0.108	coll t=2	0.024
	Rs t=2-3	0.132	Rs t=3-4	0.140
	Rs t=4-5	0.597		
Landform	C16	0.028	C22	0.043
	C23	0.036	D32	0.082
	D34	0.022	D37	0.069
	D38	0.107		
Land Use	MTC[1]	0.005	Paddy	0.002
	AC[2]	1.524	SL[3]	0.053
	Rubber	0.006	Tea	0.052
	DF[4]	0.051	DMF[5]	0.058
	VH[6]	10.583	Rubber/SL[3]	1.253
Hydrology	0-300	1.43	300-600	2.62
	600-900	2.47	900-1200	2.21
	1200-1500	1.86	1500 <	0.39
Rainfall	303-439	2.05	439-571	1.24
	571-704	0.71	704-836	0.27
	836-969	0.0	969-1071	0.0
Slope	18-31	0.46	32-40	1.28
	41-66	1.74	66-80	2.31
Aspect	143-215	0.57	215-287	1.48
	287-360	2.98		
SPI	665-885	1.34	886-1191	1.52
	1192-1546	2.05		
TWI	6.8-8	0.72	8.0-10.0	1.42
	10.0-12.0	1.81	12.0 <	2.05
STI	5.0-7.58	0.61	7.5-9.0	1.33
	9.0-11.5	1.52	11.5 <	2.33

TABLE I
FREQUENCY RATIO OF CONDITIONING FACTORS

[1] MTC:Mixed Tree Crops [2] AC:Annual Crops [3] SL:Scrub Land
[4] DF:Degraded Forest [5] DMF:Dense Mixed Forests
[6] VH:Village Home-lets

Slope angle and aspect varying between 66°-80°and 287-360 respectively show the highest FR values demonstrating increased susceptibility. It can also be seen that SPI between 1192-1546, TWI greater than 12 and STI greater than 11.5 are having the highest FR values indicating high susceptibility to landslides.

The 12 factors that were initially considered, were subjected to an analysis using Information Gain Ratio, to filter the most reliable set of conditioning factors. Table II shows the calculated entropy values, Information Gain, Split Information Gain values and Information Gain Ratios for the selected 12 parameters.

The preliminary results showed that the highest information gain ratio is for soil thickness(0.1568). It is followed by geology(0.1456), landform(0.101), land use(0.025), and soil type(0.0035). Since none of the conditioning factors indicated null predictiveness to the landslide occurrence, all 12 factors are considered in the construction of the machine learning model using Random Forest.

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Conditioning Factor	Info(S)	Info(S,A)	Split.Info(S,A)	Info.Gain(S,A)
Geology	0.1562	0.0272	0.8858	0.1456
Soil Thickness	0.1562	0.0418	0.7294	0.1568
Soil Type	0.1562	0.0964	1.7069	0.0035
Landform	0.1562	1.5762	14.5067	0.1010
Land use	0.1562	0.0941	2.4816	0.0250
Hydrology	0.1562	0.0241	1.8320	0.0721
Rainfall	0.1562	0.0035	2.3333	0.0654
Slope	0.1562	0.0121	2.341	0.0615
Aspect	0.1562	0.0881	1.8765	0.0362
SPI	0.1562	0.0144	1.0916	0.1299
TWI	0.1562	0.0552	1.7336	0.0582
STI	0.1562	0.1103	2.4816	0.0184

TABLE II
INFORMATION GAIN RATIO - TERRAIN FACTORS

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